

Product Requirement Document (PRD)

1. Executive Summary & Objectives

IRVision AI is a satellite-based remote sensing enhancement platform designed to process, upscale, and colorize grayscale infrared (IR) imagery. Infrared imagery captures thermal emissions from the Earth's surface and clouds, but scientists often struggle to visually isolate temperature gradients in raw grayscale format. The objective of this project is to provide a real-time, interactive, secure web system that performs automated multi-spectral pseudo-color mapping and super-resolution upscaling, aiding scientists in meteorology, oceanography, and environmental monitoring.

2. Target User Personas

- **Meteorological Analyst:** Needs to inspect storm eye formations, cloud top temperatures, and wind structures to track hurricane developments.
- **Oceanographer:** Analyzes sea surface temperature (SST) gradients to identify marine currents, thermal fronts, and upwelling zones.
- **Remote Sensing Scientist:** Requires high-resolution, colorized datasets with clear spatial features for land classification studies.

3. Core Functional Requirements

Requirement ID	Feature Group	Description	Priority
FR-101	Secure Auth & SSO	Support JWT authentication and standard logins alongside Google OAuth 2.0 Single Sign-On.	High
FR-102	Forgot Password	Enable multi-step password recovery using 6-digit OTP verification codes sent via SMTP.	High
FR-103	Pipeline Processing	Execute noise filtering, contrast equalization, super-resolution, and colorization.	High

FR-104	Progress SSE	Provide a real-time Server-Sent Events progress stream connection during upload.	Medium
FR-105	Interactive Viewers	Offer slider compare, zoom magnifiers, and triple-grid result layouts.	Medium
FR-106	History Log	Maintain a downloadable historical log of processed images with search/filter tools.	Medium

4. Non-Functional Requirements

- **Performance:** Image enhancement and colorization pipeline must execute in under 2 seconds per file (tested at 0.1s on lightweight CPU mode).
- **Memory Constraint:** The server must operate safely under 512MB RAM without triggering hosting OOM crashes.
- **SEO and Title Guidelines:** Static routes must contain descriptive HTML metadata titles and semantic header hierarchies.